Assignment 2

Please work on the following questions. Write your answers and paste your charts in a separate word file. (It is preferable that you generate the charts in tableau/power BI)

1. (30 pts) **Busiest North American Airports.** Based on the total passenger traffic, the airports in the file ***Airports*** ***from Chapter 5 in dataset.zip*** are among the busiest in North America.
    a. Construct a **histogram of the frequency counts** using a bin width of 10 with the first bin starting at a value of 30 million.
    b. What is the most common passenger traffic range based on this histogram?
    c. Describe the shape of the histogram.

2. (40 pts) **Missing Data for Product Satisfaction Rating.** To understand the implications of missing data, we must explore the patterns associated the missing entries for the Product Satisfaction Rating variable in the ***EJB*** ***from Chapter 6 in dataset.zip***.
    a. Construct the **relative(percent) frequency distribution** of records over the values of the **Product Satisfaction Rating** variable. What percentage of records are missing a value of Product Satisfaction Rating?
    b. Considering only records that **report values** of **Product Satisfaction Rating**, construct the **relative(percent) frequency distribution** of records over the **different flavors**.
    c. Considering records that are **missing values** of **Product Satisfaction Rating,** construct the **relative(percent) frequency distribution** of records over the **different flavors**.
    d. Compare the distributions in part (b) and part (c). What does this comparison suggest?

3. (30 pts) **Market Capitalization and Profit By Sector.** The file *Fortune500Sector in chapter 6* contains data on the profits, market capitalizations, and industry sector for a recent sample of firms in the ***Fortune500Sector*** ***from Chapter 6 in dataset.zip***.
    a. Differentiating observations by using a **different color** for **each industry sector**, prepare **a scatter chart** to show the relationship between the variables Market Capitalization and Profit in which Market Capitalization is on the vertical axis and Profit is on the horizontal axis.
    b. Emphasize the relationship between Market Capitalization and Profit within the **Industrials sector** by formatting all other sectors with data points in gray with no fill. Create a **trendline** based only on the observations in the **industrial** sector. What does the trendline indicate about this relationship between Market Capitalization and Profit within the industrials sector?
    c. Construct 95% confidence intervals for question (b) above. (click your trendline→Edit→Options→check "Show confidence bands" to create confidence intervals)

General Submission Instructions:
All work must be your own. Copying other people's work or from the Internet is a form of plagiarism and will be prosecuted as such.

You may submit a Microsoft Word (.doc/.docx) document as an attachment using the Canvas Assignment tool, or you may copy and paste your answer into the provided box within the Assignment tool. If you attach a document for your assignment, be sure to include your name in the text of the document and in the name of the document.

- You can only submit once, so make sure you are completely finished before submitting and that you attach the correct word .doc/.docx file.
- Submissions sent by email will NOT be accepted.

Due dates are listed in the Assignment Schedule document.